Strength Prediction and Sensitivity Analysis of Geopolymer Concrete Using Soft Computing Techniques

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- Sensitivity analysis.

Abstract: Researchers have been looking for alternative materials in the concrete manufacturing sector due to the need for sustainable materials that would lower the quantity of CO_2 produced during the manufacture of Ordinary Portland cement (OPC). Industrial by-products such as fly ash and slag, are beneficial in the production of Geopolymer concrete (GPC), which has become a suitable substitute for OPC concrete due to its better strength, temperature stability, a stronger bond, and chemical resistance. In this study, AI-based machine learning Algorithms (MLA) and artificial neural networks (ANN) models are effectively adopted for predicting the compressive strength of supplementary cementitious materials (SCM) based GPC. The model's performances were assessed using the coefficient of determination R^2, RMSE, NSE, and d. Compared among the models, the ANN5 (2 hidden layers with 80 & 70 neurons) and MLA (i.e., Optimizable Ensemble) models had lower values of the errors RMSE, as well as higher values of R^2, NSE, and d, which indicates the models' better performance. To analyze the effect of each parameter, sensitivity analysis was performed. It was seen that the compressive strength of GPC is strongly affected by hydration age and slag contents. However, the outcomes of this study might be applied to predict the mechanical characteristics of SCMs-based GPC that will assist the field of civil engineering by saving time and optimum design of GPC.

1. Introduction

Researchers have been interested in cementitious materials made from geopolymers because of the climatic changes brought on by global warming caused by greenhouse gas emissions from the manufacture of cement (Jindal, 2019). Geopolymer concrete's (GPC) mechanical qualities are now being improved by researchers in an ongoing effort to increase its potential for use in building construction. Geopolymer concrete is an eco-friendly and cost-effective building material that is created by the chemical reaction of inorganic molecules and uses industrial by-products as its raw material. By reacting a solid aluminosilicate powder with alkali hydroxide/alkali silicate, geopolymers are created (Davidovits, 1991). Slag and fly ash, which are byproducts of coal produced from thermal power plants, are widely accessible everywhere. Fly ash can be activated with sodium metasilicate (Na_2SiO_3) to create a geopolymer substance that acts as a binder. A transparent byproduct of the melting of iron ore is called blast furnace slag. It consists of silicon dioxide mixed with [steel] oxide. As an OPC substitute, slag improves performance, decreases

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lifecycle costs, and also boosts compressive strength. In place of the current plain cement concrete, it is the most climate-friendly substitute construction material for a sustainable future.

Compressive strength (CS) is the most important aspect to consider while designing and analyzing concrete. Artificial intelligence (AI) has made great progress in the field of civil engineering, especially when it comes to predicting the behavior and performance of materials like concrete (Khan et al., 2021). Various machine learning techniques are frequently employed to anticipate desired outcomes. These machine-learning techniques can forecast concrete's mechanical properties with accuracy. A variety of technological and scientific problems may be solved using artificial neural networks. It is commonly used to develop complex, basically nonlinear processes' predictive statistical models. In keeping with the above-mentioned, the great need for predicting the compressive strength of geopolymer concrete using progressed machine learning techniques is taken into consideration. In this study, the potential application of different soft computing models, i.e., Artificial Neural Networks (ANN), and Machine learning algorithms (MLA) are inspected for determining the compressive strength of GP concrete based on fly ash (Anam Siddique et al., 2011).

The aims of the study were: (1) to analyze the strength prediction performance of geopolymer using artificial intelligence techniques such as ANN and MLA; (2) to evaluate the best fit model using statistical parameters such as R^2, RMSE, NSE, and d; finally, (3) sensitivity analysis to find out the influence of input parameters to model output.

2. Methods of Analysis

2.1. Source of data

In this research, a total of 1000 data sets were gathered from the published papers (Ahmad et al., 2021, 2022; Assi et al., 2016; Dong et al., 2021; Farooq et al., 2021; Ghafoor et al., 2021; Haruna et al., 2021; Huynh et al., 2020; Ishak et al., 2019; Kishore et al., 2022; Lokuge et al., 2018; Nath & Sarker, 2014, 2017; Sarker et al., 2013; Shi et al., 2012; van Dao et al., 2019; Vora & Dave, 2013; Wardhono et al., 2017; Yadollahi et al., 2015; Zhang et al., 2020) to produce data for running machine learning-based models. The present study designated the CS of GPC as the output variable. The Fly Ash, Slag, Na_2SiO_3 (Sodium Metasilicate), Fine Aggregate, Coarse Aggregate, Water, and Age was taken as input variables for building the models to predict the Compressive Strength (CS). The data points were divided into three sets such as training (train), testing (test), and validation (valid), where 70% of data were selected randomly for training the adopted models, 15% for testing, and the other 15% for validation. Each group of data was utilized throughout the analysis by training, testing, and validation to figure out a better predictive model with high accuracy to the experimental observations of output variables. Descript statistics were also carried out to understand the nature of the collected data more precisely which is shown in Table 1. The purpose of a descriptive statistic is to summarize data and it can be useful to provide basic information about variables in a dataset and to highlight potential relationships between variables.

Table 1: Descript statistics of input and output variables of the data set.

Types	Input Parameters (Kg/m^3)							Output
	Fly Ash	Slag	Na_2SiO_3	Water	Fine	Coarse	Age (day)	CS (MPa)
Mean	0.32	0.27	0.13	0.48	0.46	0.50	0.20	35.55
Standard Error	0.01	0.01	0.00	0.00	0.01	0.01	0.00	0.52
Median	0.10	0.15	0.13	0.50	0.47	0.49	0.16	33.87
Mode	0.10	0.10	0.12	0.55	0.42	0.40	0.16	33.40
Standard Deviation	0.26	0.19	0.06	0.14	0.16	0.18	0.14	16.57
Sample Variance	0.07	0.04	0.00	0.02	0.03	0.03	0.02	274.49
Kurtosis	-1.37	-0.57	95.60	0.16	-0.04	-0.62	12.10	-0.28
Skewness	0.49	0.76	9.29	0.07	-0.25	-0.04	3.27	0.43
Range	0.80	0.80	0.80	0.80	0.80	0.80	0.80	80.27
Minimum	0.10	0.10	0.10	0.10	0.10	0.10	0.10	2.33
Maximum	0.90	0.90	0.90	0.90	0.90	0.90	0.90	82.60
Sum	323.14	269.42	133.45	480.58	462.63	498.36	198.27	35,551.22
Count	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000

2.2. Employed AI Models

2.2.1 Artificial Neural Network

The ability of artificial neural networks to learn from the display of sample data sets (patterns) is one of its most important characteristics. This is performed by delivering a variety of training samples to the network. The feed-forward neural network with multilayer perception is a frequently used ANN design. The artificial neural network (ANN) is one of the models that are most frequently employed in the domains of machine learning and data mining. Inspired by biological neural networks in human brains, this model is a network of linked nodes that may be trained to predict the expected output from the input provided (Narayanakumar & Raja, 2016) This technique offers significant benefits not seen in conventional computational models since it does not need assumptions or preconceived limits about the structure of the model for example, between dependent and independent variables (Naik et al., 2020).

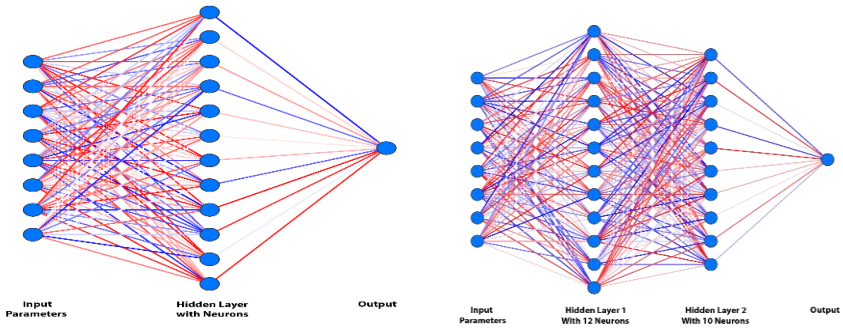


Figure 1. The architecture of ANN for eight input parameters, one hidden layer with twelve neurons, and eight input parameters, two hidden layers with twelve neurons.

2.2.2. Optimizable Machine Learning Algorithm

Machine Learning Algorithms are a type of data analysis that automates the creation of analytical models. It's a field of artificial intelligence based on the premise that computers can learn from data, recognize patterns, and make judgments with little or no human input. The following MLA models were applied in this study to predict the CS of GPC:

Optimizable Decision Tree (ODT)

An Optimizable Tree is a well-known approach that uses a tree assembly to represent decisions and likely outcomes from data. A deeper tree contains more sub-trees, resulting in more precise conclusions. Furthermore, establishing a calculating function, which is indicated by 'criterion' in sklearn, elevates the quality of splits.

Optimizable Support Vector Machine (OSVM)

In general, OSVM methods are based on mapping data points from low-dimensional space to high-dimensional space to assemble data that is linearly divisible, a phenomenon that is employed for both classification and regression problems. The categorization margin is then transferred to the divider data points via a hyperplane. OSVM's objective function for n data points is:

$$\arg \min_{w} \left\{ \frac{1}{n} \sum_{i=1}^{n} \max\{0, 1 - y_i f(x_i)\} + CW^T W \right\} \qquad \text{Eq. 1}$$

Here, W is the normalization vector and C denotes the penalty parameter of the error term, which is an important hyper-parameter in OSVM simulations. In all OSVM representations, the kernel function f(x) is used to quantify the connection between two data points xi and xi, and it is favored over various types of kernels. As a result, the kernel nature would be an active hyper-parameter that could be tweaked. Linear, polynomial, radial-based, and sigmoid-based kernels are the most well-known kernel functions in OSVM.

Optimizable Gaussian Process Regressor (OGPR)

The Gaussian Process Regressor (OGPR) is a subset of the MLA method. It is built based on the Gaussian stochastic approach and the Bayesian learning notion. The accuracy of OGPR models is largely determined by how well the model hyperparameters work. If a set of N observations of the input-target pairings D = xn, yn Nn = 1 is used, the projected average and co-variance in an OGPR model may be expressed as-

$$\mu(x;\{x_n,y_n\},\theta) = K(X,x)^T k(X,X)^{-1}\big(y - m(X)\big) \qquad \text{Eq. 2}$$

$$\sum(x,x';\{x_n,y_n\},\theta) = K(x,x') - K(X,x)^T K(X,X)^{-1} K(X,x') \qquad \text{Eq. 3}$$

Here, θ = GP hyperparameters and K (X, x) = N-dimensional column vector of cross-covariance amid x as well as the set X. Also, The N×N matrix K (X, X) = Gram matrix for the set X. In addition, the acquisition function is used to adapt the expected improvement (EI), which is written as-

$$aEI(x;D,\theta) = \sigma(x;D,\theta)\big[\gamma(x)\varphi\big(\gamma(x)\big) + N(\gamma(x);0,1)\big] \qquad \text{Eq. 4}$$

Here, $\gamma(x) = \frac{f(x_{best} - \mu(x;D,\theta)}{\sigma(x;D,\theta)}$; f ($X_{best}$) indicates the lowermost observed value and ∅(·) presents the collective distribution function; μ (x; D, θ) = predictive mean function and σ (x; D, θ) = predictive variance function. Furthermore, the best current value is denoted by- $X_{best} = arg_w^{min} f(X_n)$.

Optimizable Ensemble (OER)

The observed data of CS were forecasted using optimizable MLA models such as OTR, OSVM, OGPR, and OER. The model performance was improved by integrating numerous decision trees using ensemble algorithm-based regression models. Using the bagging methods, the ensemble is made up of the 'base estimator' and 'n estimators'.

2.3. Model Evaluation Criteria

The prediction performance of nominated ANN and optimizable MLA models were assessed using the well-known goodness of fit statistical parameters namely correlation coefficient (`r), root mean square error (RMSE), Nash Sutcliffe model Efficiency coefficient (NSE), and index of agreement (d). The mathematical expression and typical range of good fitness of these performance indicators are presented in Table 2. The correlation coefficient is a statistical measure of the strength of the relationship between the relative movements of your two variables. The values range between -1.0 and 1.0. The RMSE measures how much error there is between two data sets. It is a frequently used measure of the difference between a predicted value and an observed or known value. The NSE is a normalized statistic that determines the relative magnitude of the residual variance compared to the measured data. The d is a measure of the degree of model prediction error. It can detect additive and proportional differences in the observed and simulated means and variances.

Table 2: Goodness of fit parameters for model evaluation

Expression of parameters	Standard Range
$\mathbf{r} = \frac{n(\sum XY) - (\sum X)(\sum Y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$ Here, r = Pearson correlation; n = number in the given dataset; x = first variable in the context (or observation data); y = second variable (or modeled data)	0 - 1
$\boldsymbol{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(X_{obs,i} - X_{model,i}\right)^2}{n}}$ Here, X (Obs, i) is the value of the observation, and X (model, i) is the value of the forecast.	0 is best
$\boldsymbol{NSE} = 1 - \frac{\sum_{i=1}^{n}(OBS_i - SIMi)^2}{\sum_{i=1}^{n}(OBSi - \overline{OBS})^2}$ Here, OBSi is the observation value and SIMi is the forecast value and OBS-bar is the average of observation values.	0 - 1
$\boldsymbol{d} = 1 - \frac{\sum_{i=1}^{n}(O_i - P_i)^2}{\sum_{i=1}^{n}(\lvert P_i - \bar{O}\rvert + \lvert O_i - \bar{O}\rvert)^2}$ Here, Oi is the observation value and Pi is the forecast value and Obar is the average observation value and Pbar is the average forecast value.	0 - 1

2.4. Sensitivity Analysis

Sensitivity analysis (SA) determines the relative contribution of the input variables utilized to predict the compressive strength of GPC. SA explains how the input variable influences the result. To quantify the effect of each input variable on the compressive strength of GPC, a sensitivity analysis was done. The greater the absolute value of any input parameter, the more important it is CS, and the lower the absolute r-value. However, in some circumstances, determining the positive and negative effect of input parameters on output variables requires more than the absolute factor. As a result, the r-value with directionality was utilized in this study since it allowed for a more intuitive explanation.

$$\boldsymbol{r(I_j, O)} = \frac{\sum_{i=1}^{N}(\boldsymbol{I_{j,i}} - \bar{\boldsymbol{I}}_j)(\boldsymbol{O_i} - \bar{\boldsymbol{O}})}{\sqrt{\sum_{i=1}^{N}(\boldsymbol{I_{j,i}} - \bar{\boldsymbol{I}}_j)^2 \sum_{i=1}^{N}(\boldsymbol{O_i} - \bar{\boldsymbol{O}})^2}} \qquad \text{Eq. 1}$$

Here, $I_{j,i}$ and $\bar{I}_j$ are the ith and mean value of i^{ts} input variables respectively, and $O_{j,i}$ and O are the ith and mean values of predicted out variables by the proposed mathematical model.

3. Result and Discussions

3.1 Prediction performance of ANN models

The AI-based Artificial Neural Networks (ANN) were effectively adopted for predicting the compressive strength (CS) of geopolymer concrete. The model's performance was assessed using the coefficient of determination R^2, RMSE, NSE, and d. The proposed models with one- and two-hidden-layer demonstrated improved calibration to outputs with high R^2, NSE, and d closer to 1, as well as lesser RMSE, indicating a higher fitness in training and testing data sets but the study discovered that the ANN5 model, which includes two hidden layers with 80 and 70 neurons is the best-fitted model because it has higher R^2, NSE, and d with least error for the training, testing, and validation data points compare to other models.

Table 3: Evaluation of the representative ANN models.

SL	HL	N		R^2			RMSE			NSE			d		
				Train	Test	Valid	Train	Test	Valid	Train	Test	Valid	Train	Test	Valid
ANN1	1	10		0.89	0.85	0.86	5.55	6.39	6.45	0.89	0.84	0.86	0.97	0.95	0.96
ANN2	1	70		0.96	0.87	0.87	3.43	6.37	5.59	0.96	0.87	0.87	0.99	0.96	0.97
ANN3	1	100		0.96	0.84	0.83	3.31	6.24	7.06	0.96	0.83	0.82	0.99	0.95	0.95
ANN4	2	60	50	0.96	0.81	0.87	3.14	7.33	6.63	0.96	0.80	0.87	0.99	0.95	0.96
ANN5	2	80	70	0.94	0.90	0.87	4.08	5.22	6.16	0.94	0.90	0.87	0.98	0.97	0.96
ANN6	2	100	90	0.96	0.71	0.83	4.68	11.03	7.89	0.92	0.55	0.73	0.98	0.90	0.94

3.3 Prediction performance of optimizable MLA models

Optimizable hyperparameter-based MLA models were tested for predicting compressive strength (CS) for training, testing, and validation data sets. For seven input parameters and one output parameter, there was no significant difference in model assessment statistics or prediction outputs among the assigned MLA models. The OER model has a higher value of R^2, NSE, and d for the training, testing and validation data set compared to the other three MLA models and also has a lower value of RMSE. This indicates that OER is the best-fitted model compared to other MLA models. Table 4 shows the optimized Optimizable Ensemble hyperparameters for seven inputs, where the hyperparameters of Optimizable Ensemble were improved using a Bayesian optimization technique over 30 iterations. Optimizable Ensemble model by expressing the quite aggregation of observed versus prediction values of Optimizable Ensemble along the linear fit line with R^2, RMSE, NSE, and d for CS.

Table 4. Evaluation of the representative Optimizable MLA models.

Models	R^2			RMSE			NSE			d		
	Train	Test	Valid	Train	Test	Valid	Train	Test	Valid	Train	Test	Valid
ODT	0.92	0.78	0.75	4.82	8.32	8.56	0.92	0.75	0.74	0.98	0.94	0.93
OSVM	0.79	0.83	0.74	7.66	6.78	8.63	0.78	0.83	0.74	0.94	0.95	0.92
OGPR	0.97	0.94	0.92	2.98	4.21	4.84	0.97	0.94	0.92	0.99	0.98	0.98
OER	0.99	0.95	0.91	1.53	3.70	4.99	0.99	0.95	0.91	1.00	0.99	0.98

3.4 Well-fitted ANN and optimizable MLA models

The efficiency of the developed models was evaluated using five different statistical parameters: R^2, RMSE, NSE, and d. Table 5 shows the statistical comparison of ANN5 and Optimizable Ensemble models for predicting the performance of experimental CS data points of GPC composite materials which shows that OER is more suited to actual CS than ANN5 because it has a higher value of R^2, NSE, and d with lest RMSE value. Figure (2-4) also provides a comparison of model predictions of compressive strength of GPC mixtures of training, testing, and validation data points which also indicates the best fitness of the OER model.

Table 5: Statistical comparison of ANN5 and Optimizable Ensemble (OER) models to whole CS data sets with predicted results.

MLA Models	R^2			RMSE			NSE			d		
	Train	Test	Valid	Train	Test	Valid	Train	Test	Valid	Train	Test	Valid
ANN5	0.94	0.90	0.87	4.08	5.22	6.16	0.94	0.90	0.87	0.98	0.97	0.96
Optimizable Ensemble	0.99	0.95	0.91	1.53	3.70	4.99	0.99	0.95	0.91	1.00	0.99	0.98

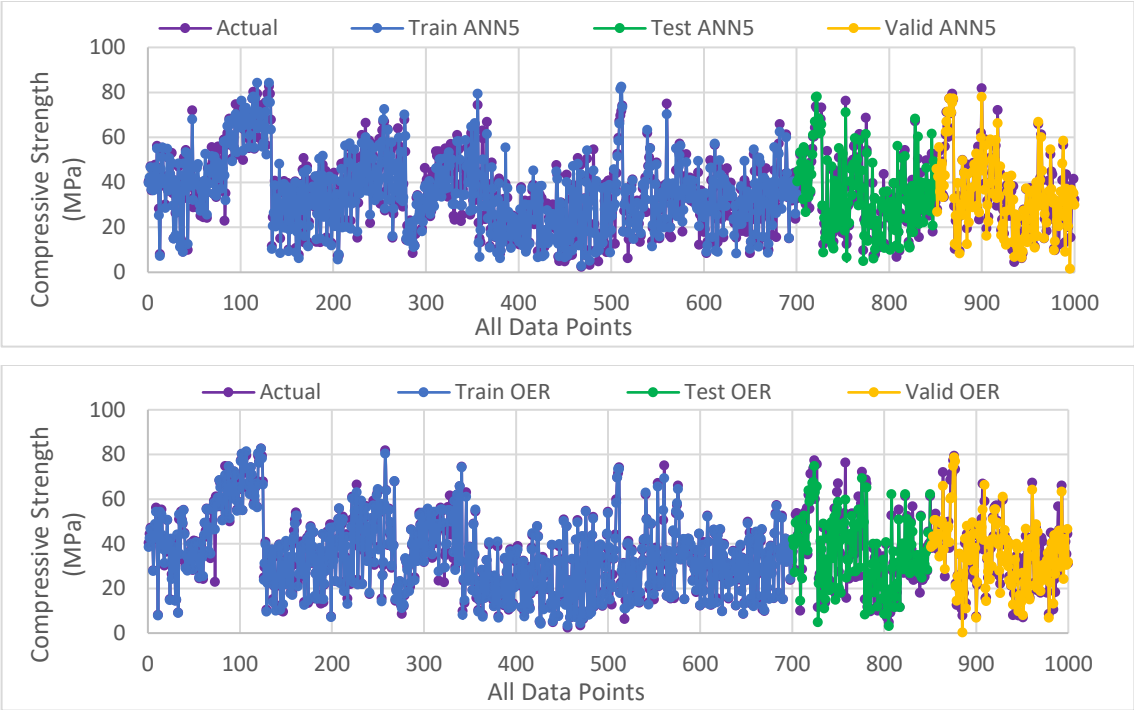
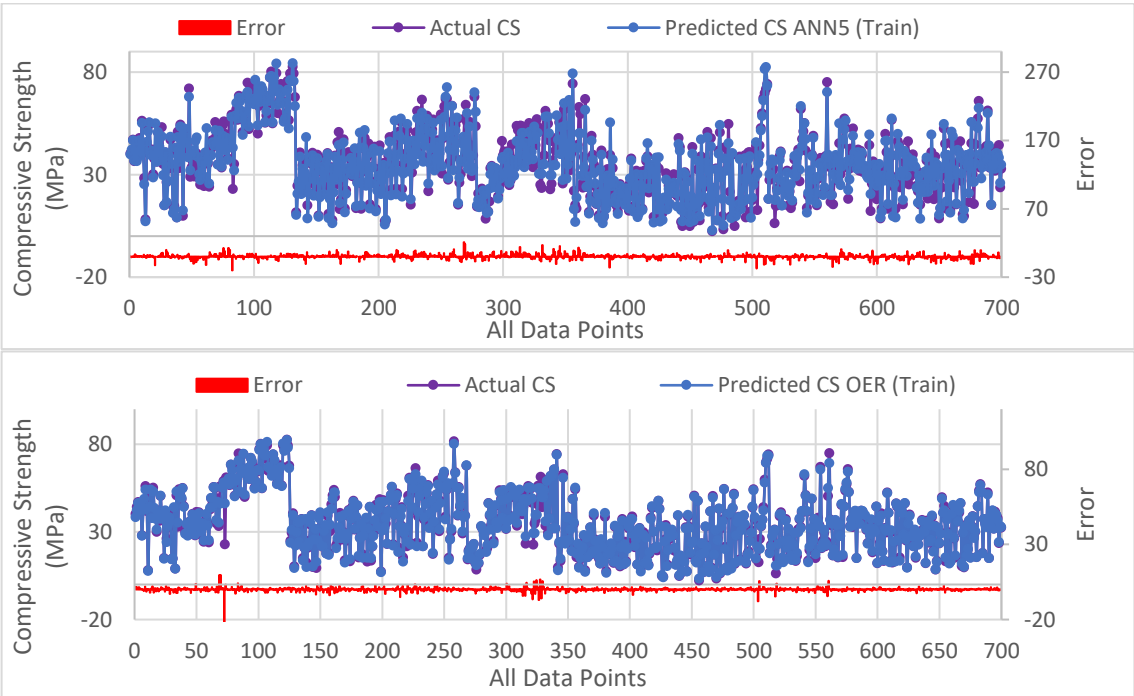


Figure 2. Scatter graph of fitness by actual CS with the predicted CS of ANN5 and OER.



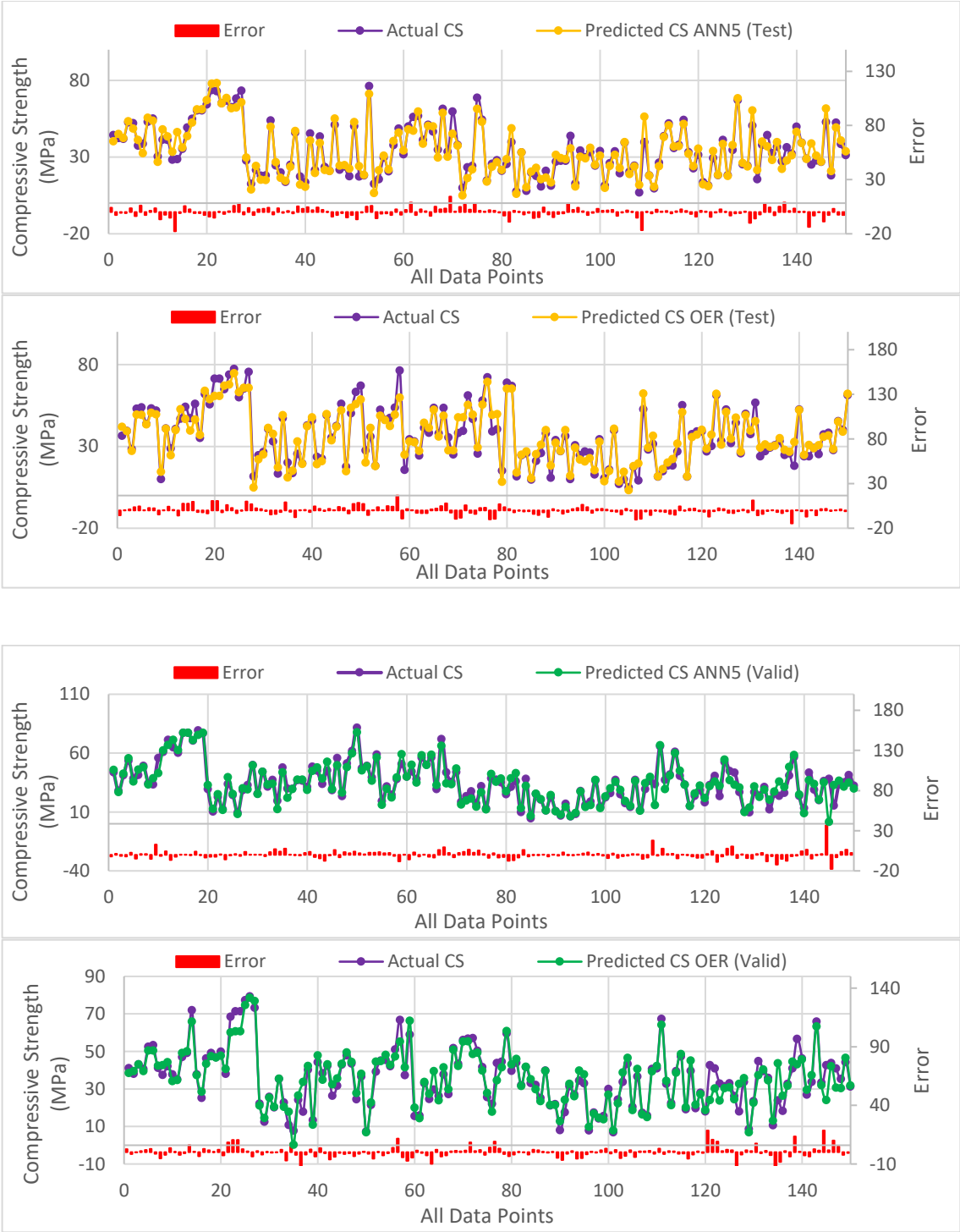


Figure 3. Calibration of ANN5 and OER models to whole observed CS data points at training, testing, and validation phases.

Figure 4. Comparison of ANN5, and Optimizable Ensemble models with prediction bounds.

7. Sensitivity Analysis

Sensitivity analysis was carried out to assess the relative contribution of input factors used to estimate the compressive strength of GPC formed using FA. As is observed, Slag and Age are found as the most important variables in the MLA and ANN models for predicting the compressing strength of GPC. Other input parameters like Fly Ash, Na_2SiO_3, Coarse Aggregate, and Water have an inverse proportional relation with the predicted compressed strength. Table 6 illustrates the results of the Sensitivity analysis.

Table 6. Sensitivity analysis of ANN5 and OER models:

Data Type	Model	Fly Ash	Slag	Na_2SiO_3	Water	Fine	Coarse	Age
All data	ANN5	-0.03	0.12	-0.15	-0.30	-0.14	-0.25	0.37
	OER	-0.09	0.14	-0.08	-0.28	-0.16	-0.17	0.34
Training	ANN5	-0.13	0.15	-0.09	-0.29	-0.14	-0.19	0.35
	OER	-0.09	0.14	-0.08	-0.28	-0.16	-0.17	0.34
Testing	ANN5	-0.14	0.16	-0.12	-0.27	0.01	-0.38	0.47
	OER	-0.12	0.18	-0.18	-0.21	-0.02	-0.41	0.48
Validation	ANN5	-0.12	0.19	-0.18	-0.33	-0.19	-0.09	0.30
	OER	-0.05	0.23	-0.17	-0.37	-0.26	-0.11	0.28

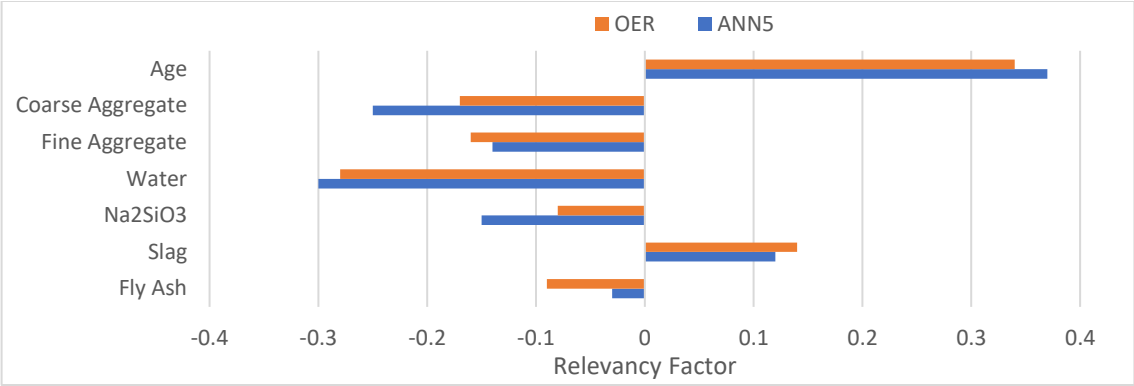


Figure 5. Relevancy factor of input parameters to output parameter according to sensitivity analysis results.

4. Conclusion

The use of a reliable and accurate model to predict the compressive strength of FA-GPC can save both time and cost. In this study, ANN and MLA are used to create a mathematical formula for the compressive strength of geopolymer concrete that is dependent on FA and GPC. Compressive strength is the most important aspect to consider while designing and analyzing concrete. Predictive models for the GPC were proposed using ANN and MLA models in this work. The conclusion can be formed based on the 1000 gathered datasets from previous research studies. The statistical measure considered in this study is R^2, RMSE, NSE, and d.

- The Optimizable Ensemble model has better performance than the other optimizable MLA model.
- The predicted result of an ANN model with one and two hidden layers was comparable to the Optimizable Ensemble model. The ANN5 (2 hidden layers and 80-70 Neurons) provided the best-predicted values for the training, testing, and validating datasets.
- The designated ANN5 and OER models could be successfully used to develop predictive models for the compressive strength of GPC.
- The assessment and comparison of statistical parameters R^2, RMSE, NSE, and d, for all the training, testing, and validating datasets validate the accuracy of the developed models properly.
- According to the sensitivity analysis approaches, the slag and age content are the most effective input variables for predicting the compressive strength of fly ash-based GPC for the models.

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